Statics – Resolving Forces A

A-Level Physics

For each question, draw a force diagram and resolve the forces horizontally and vertically.

Where required, take $g = 9.81\,\text{m}\,\text{s}^{-2}$.

1. A 20 kg box rests on a horizontal floor. A rope pulls the box with a tension of 75 N at 30° above the horizontal.

   Calculate the normal reaction force exerted by the floor.

2. A 35 kg crate rests on a horizontal floor. A force of 120 N acts at 25° below the horizontal.

   Calculate the normal reaction force exerted by the floor.

3. A box remains stationary on a horizontal floor. It is pulled by a rope with a tension of 140 N acting at 40° above the horizontal.

   Calculate the frictional force acting on the box.

4. A 22 kg box rests on a horizontal floor. A rope pulls it with a tension of 90 N at 35° above the horizontal. The box remains stationary.

   Calculate:

   (a) the normal reaction force from the floor;

   (b) the frictional force acting on the box.

5. A 40 kg crate rests on a horizontal floor. A force of 160 N is applied at 20° below the horizontal. The crate remains stationary.

   Calculate:

   (a) the normal reaction force;

   (b) the frictional force.

6. A 12 kg object is held stationary against a vertical wall by a force of 200 N acting at 40° above the horizontal towards the wall.

   Calculate:

   (a) the normal reaction force exerted by the wall;

   (b) the frictional force acting between the object and the wall.

7. A 25 kg box rests on a horizontal floor. A force of 180 N acts at 50° below the horizontal to the right. A second force acts horizontally to the left and keeps the box in equilibrium.

   Calculate:

   (a) the horizontal force acting to the left;

   (b) the normal reaction force from the floor.

8. A 30 kg crate rests on a horizontal floor. It is pulled by a cable with a tension of 150 N acting at 35° above the horizontal to the right. A horizontal force acts to the left and keeps the crate stationary.

   Calculate:

   (a) the horizontal force acting to the left;

   (b) the normal reaction force from the floor.

9. A 24 kg box rests on a horizontal floor. A rope pulls the box with a tension of 100 N. The box remains stationary and the normal reaction force from the floor is 190 N.

   Calculate:

   (a) the vertical component of the tension;

   (b) the angle that the rope makes above the horizontal.

10. A 30 kg crate rests on a horizontal floor. A force of 140 N acts at an unknown angle below the horizontal. The normal reaction force from the floor is 370 N.

    Calculate:

    (a) the vertical component of the applied force;

    (b) the angle that the force makes below the horizontal.

Statics – Answer Key

1. Normal reaction force: 158.7 N

2. Normal reaction force: 394.0 N

3. Frictional force: 107.2 N

4. Normal reaction force: 164.2 N
   Frictional force: 73.7 N

5. Normal reaction force: 447.1 N
   Frictional force: 150.4 N

6. Normal reaction force: 153.2 N
   Frictional force: 10.8 N downward

7. Horizontal force to the left: 115.7 N
   Normal reaction force: 383.1 N

8. Horizontal force to the left: 122.9 N
   Normal reaction force: 208.3 N

9. Vertical component of tension: 45.4 N
   Angle above the horizontal: 27.0°

10. Vertical component of applied force: 75.7 N
    Angle below the horizontal: 32.7°